

OA3103: Data Analysis

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Lectures/Lab: GL-118

Office Hours: By appointment
Meeting Times: M-F 0900-0950

Purpose: Serve as a **statistical modeling** bridge between OA3102 and OA4106/OA4118. Develop the modeling intuition, computational fluency, and analytic workflow that are needed to interrogate real data and build defensible, reproducible analyses.

Course Goals:

- **Develop a reproducible, Python/R-based analytic workflow**
 - Together we will learn to structure projects, manage data, write clean code, and use Python/R pipelines to standardize preprocessing and modeling.
- **Build the statistical modeling backbone needed for OA4106 and OA4118**
 - Students will master multiple linear regression, GLMs, validation, cross-validation, PCA (light), and introductory classification and clustering.
- **Cultivate conceptual understanding (Develop our *theoretical tools*)**
 - Students will understand the modeling assumptions, loss functions, and statistical logic behind the tools they use — with light derivations to support our intuition.
- **Gain hands-on experience with realistic datasets (Develop our *coding tools*)**
 - Weekly labs and assignments will build a cumulative “Cookbook for Data Analysis and Statistical Modeling.”
- **Introduce concepts that will be deepened in 4106 and 4118**
 - Students will ideally see concepts multiple times at increasing depth.
- **Prepare students to analyze and brief real DoD-relevant data**
 - Students will practice communicating uncertainty, limitations, and operational relevance.

Textbooks/Resources.

- **ISLP:** *An Introduction to Statistical Learning with Applications in Python*, Gareth James et al.
- **Shalizi:** *Advanced Data Analysis from an Elementary Point of View*, Cosma Rohilla Shalizi
- **EDA:** *Introductory Statistics with Armed Forces and Defense Applications* (Chapter 3: Exploratory Data Analysis), Robert Bassett
- **sklearn:** <https://scikit-learn.org/stable/>
- **Faraway:** *Linear Models with R 2nd Edition*, Julian J. Faraway

With the exception of Faraway, these resources are freely available via the links in the Sakai Site. Additionally, we may occasionally reference the textbook *Probability and Statistics for*

Engineering and the Sciences (9th ed.) by Jay L. Devore that you used last quarter. I will provide summaries or excerpts from Faraway and Devore as needed.

Weekly Schedule. 4 Lectures plus 1 Lab per week. The labs will be hands-on demo/do coding, setting students up for their weekly/bi-weekly assignments, which will include practical applications and some theoretical questions.

Week 1: Analyst's Workflow + Python Foundations

- *30 Mar – 3 Apr*
- Lecture themes: What is data analysis in the OR context; where prediction fits between inference, modeling, and machine learning; introduction to regression vs classification; reproducibility and defensibility; review of statistical concepts; workflow and basic data handling in Python and R.
- Lab: Load and clean datasets, basic EDA: simple plots and summary statistics
- Quiz 1: Directly ask questions from syllabus and introduction slides.
- Cookbook contribution: “Project setup and reproducible workflow, Basic cleaning and EDA in R / Python
- Primary reference: ISLP Chapters 1 and 2
- Additional reference: Shalizi Chapter 1 (Regression Basics)

Week 2: Exploratory Data Analysis (EDA), Intro to Linear regression

- *6 – 10 Apr*
- Lecture themes: Visual EDA; outliers/missingness/transformations; correlation structure; simple linear regression, residuals and basic diagnostics, matrix form
- Lab: seaborn pair plots, correlation heatmaps, anomaly detection, structured EDA workflow in Python and R
- Cookbook Contribution: EDA Patterns and Visual Diagnostics
- Primary reference: EDA (Bassett Ch. 3), ISLP Section 3.1

Weeks 3-4: Multiple Linear Regression (MLR) as the modeling backbone

- *13-24 Apr*
- Lecture themes: MLR formulation; interpretation of coefficients; diagnostics: residuals/leverage/influence; multicollinearity; model selection: AIC/BIC/adjusted R^2 ; prediction vs inference; LSE as MLE (when normality is assumed, conceptual seeds for 4118); optimal prediction, MSE, MAE; pipelines for preprocessing + regression; design matrices and categorical predictors (binary, dummy, one-hot encoding)
- Reading supplement: ISLP Ch. 3 (Linear Regression)
- Lab: Fit MLR with sklearn, residual and leverage plots, VIF, `cross_validate`
- Cookbook contribution: Linear Regression Modeling and Diagnostics, Regression Pipelines in sklearn and R
- Primary reference: ISLP Ch. 3 (Linear Regression)
- Additional reference: Shalizi Chapter 2

Week 5: Classification and Generalized Linear Models

- *27 Apr – 1 May*
- Lecture themes: Logistic regression; Poisson regression, when linear regression fails, iterative optimization (conceptual seed for 4118), pipelines for logistic regression, cross-entropy, confusion matrices, ROC curves
- Lab: Logistic regression with sklearn, confusion matrices, ROC curves, `LogisticRegressionCV`
- Cookbook contribution: “Logistic Regression + Classification Metrics”
- Primary reference: ISLP Ch. 4 (Classification)
- Additional reference: Shalizi Chapters 11 and 12

Midterm 1 to cover weeks 1-5

Week 6: Resampling, Uncertainty, and Validation

- *4-8 May*
- Lecture themes: Train/Test Splits; Cross-validation (k-fold, LOOCV); Bootstrap, Permutation tests, model selection logic
- Reading supplement: ISLP Ch. 5, sklearn documentation: `cross_validate`, `GridSearchCV`
- Lab: `cross_validate`, bootstrap confidence intervals, permutation test for feature importance
- Cookbook contribution: “Validation and Resampling Techniques”
- Primary reference: ISLP Ch. 5 (Resampling methods)

Week 7: PCA + Multicollinearity, Intro to Regularization

- *11-15 May*
- Lecture themes: PCA as variance decomposition; SVD intuition, PCA for multicollinearity; PCA for visualization (football); Pipelines with PCA; Scree plots, biplots; ridge regression
- Lab: `sklearn.decomposition.PCA`, scree plots, biplots, pipeline: scaling → PCA → regression
- Cookbook contribution: “PCA for diagnostics and Visualization”
- Primary reference: ISLP Ch. 12 Unsupervised Learning (PCA sections)
- Additional reference: Shalizi Ch. 15

Week 8: Classification Extensions (Statistical Framing)

- *18-22 May*
- Lecture themes: Logistic Regression Recap; k-NN; Naïve Bayes; Decision Trees; Decision boundaries; Random forests (conceptual); Evaluation metrics; sklearn/R pipelines for classification
- Lab: Fit kNN, tree, and RF models
- Cookbook contribution: “Introductory Classification Models”
- Primary reference: ISLP Ch. 4 (Classification) and Ch. 8 (Tree-Based Methods)
- Additional reference: Shalizi Chapter 13

Week 9: Clustering

- 25-29 May
- Lecture themes: k-means; hierarchical clustering; distance metrics; cluster validation; cluster interpretation, pipelines for clustering
- Lab: sklearn.cluster, dendrograms, cluster interpretation
- Cookbook contribution: “Clustering Techniques and Interpretation”

Week 10: Synthesis/Multiple testing + Review + Midterm 2

- 1-5 June
- Lecture themes: Synthesis of data analysis pipeline (regression, clustering, model selection techniques); Family-wise error rate; False discovery; Re-sampling approaches
- Lab: Multiple testing, implementing of corrections and simulation
- Primary reference: ISLP Ch. 13 (Multiple Testing)

Midterm 2 to cover Weeks 6-10

Week 11: Preview of OA4106 (No Exam/No testable topics)

- 8-12 June **Medical Appt. on 8 June:** *Expect a replacement lecture video by myself or my favorite YouTube expositors.*
- Lecture themes:
 - **Margin-based methods:** SVM
 - **Tree-based methods:** Decision Trees, Random Forests
 - **Optimization + Regularization:** LASSO, Elastic Net, Gradient Descent
 - **Representation learning:** PCA as dimensionality reduction, Neural Networks

Homework. Homework will coincide with the weekly labs and will consist of theoretical questions and a coding application. It will be typically collected via Sakai by the day of the following lab session. It will primarily be graded for completion (read: you are responsible for using the homework as a learning reinforcement tool). However, I will spot-check one problem each week using a random generator and post the distribution/seed on Sakai.

Quizzes. Each Tuesday will include a short quiz. They will last approximately 15 minutes on Sakai and ensure you are engaging with the material from the previous week. Quizzes will open shortly after class and close by the start of class the next day. The lowest quiz score over the quarter will be dropped.

When taking quizzes, please do not rely on Sakai to be stable and sensible: submit your quiz before the time ends to make sure your answers are recorded.

Projects. There will be two projects throughout the course. Project 1 will use a prescribed dataset to reinforce concepts from the earlier part of the course. The second project will be a dataset of your choosing, subject to instructor approval. The final write up of your findings from Project 2 will be turned in by the final day of class in lieu of a final exam. Projects will be graded on the validity of the analysis, clarity of the report, and my assessment of your learning from the project.

Exams. We will have two 1-hour midterm exams, covering material from Weeks 1-4 and Weeks 6-10, respectively. Exams will emphasize the theoretical content of the course and will not include a heavy application component (read: no coding expectations). If there are any applications, they will involve some light calculation or interpretation of a data structure that I will provide.

The exams will not intentionally be cumulative; problems from Exam 2 will be specifically worded to cover concepts learned during the latter half of the course. However, the material builds on previous concepts, so you may need to rely on earlier foundational material.

You may bring a single 8½ x 11” *engineering sheet* (I loathe to call it a “cheat sheet”) on which you may include equations or formulae you think you might need. You must produce your own sheet (no sharing or asking generative AI to create it for you).

Ethics. You are adults and Commissioned/Non-Commissioned Officers or professional Civilians. I expect you to act as such.

Collaboration

- Any products you turn in represents work done by you.
- Collaboration on homework *only* is acceptable and encouraged. Anything you submit must be written and authored by you.
- You *may* ask your classmates for clarification on projects, e.g., “Did you try PCA on your dataset?” Keep in mind that there may be occasions that the *blind are leading the blind*. When in doubt, reach out to me directly via email, Teams, or in class.
- On exams and quizzes, a strict policy of *non-communication* with classmates or generative AI (see below) shall be observed.

Generative AI. Generative AI is considered a collaborative tool. **All use of Generative AI shall have a Disclosure of Generative AI statement included with your homework and projects,** in accordance with NPS policy. I support the use of Generative AI as a *tool* that supports your earning, but I will deduct points from assignments where I assess your use of AI as a *crutch*.

- Your disclosure should include:
 - What tool you used (Copilot, Claude, ChatGPT, etc.),
 - How you used it, and
 - What mitigating steps you took to ensure you were QA’ing and not over-relying on it.
- The Graduate Writing Center provides excellent guidance at <https://nps.edu/web/gwc/generative-ai>.

Improper collaboration with your classmates or GenAI will be considered as severe as plagiarism.

Grading Policy. Final grades are a weighted average of the following:

- 10% In-class participation
- 20% Homework
- 20% Quizzes
- 25% Midterm Exams
- 25% Projects

Everyone starts with full in-class participation credit. I will reach out directly if I assess any students need to be more engaged in class, are spending too much time ordering protein powder instead of paying attention to lectures (this is a true story), or if there is any other classroom conduct I deem inappropriate (see the class intro slides).

Final Disclaimers.

- This syllabus is a living document. Expect updates as necessary throughout the course of this quarter, particularly with the course schedule. Your feedback is welcome and appreciated.
- This syllabus was produced with the assistance of Microsoft Copilot to ensure clarity, consistency, and a balanced workload throughout the course schedule. To mitigate errors, the instructor drafted the syllabus, uploaded the draft for refinement, cross-checked and corrected any recommended resources, and incorporated suggestions manually rather than using any fully generated product.